Design of a Two-Bit Magnitude Comparator Based on Pass Transistor, Transmission Gate and Conventional Static CMOS Logic

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***Abstract*—Since there is a swift technological progress going on in the recent years, semiconductor industry evolved to such an extend that requirement of optimal performance in electronic circuits have become essential. Therefore, there is high requirement of energy efficient fast circuit designs in modern Integrated Circuits (IC). Comparison of two binary digits is one of the fundamental arithmetic operations in Arithmetic Logic Units (ALU) of ICs and processors. A hybrid design approach for implementing a two-bit Magnitude Comparator (MC) has been proposed in this work. The hybrid design consists of three different logic techniques namely: (a) Pass Transistor Logic (PTL), (b) Transmission Gate Logic (TGL) and (c) Conventional Static CMOS Logic (C-CMOS logic). The effectiveness of the proposed design has been compared with 6 state of art two-bit magnitude comparators using Cadence tools in 90 nm technology node. The proposed design showed best perormacne in case of delay and Power Delay Product (PDP) which proved the effectiveness of the design, Moreover, power consumption of the design is also low which makes it highly usable for portable devices that requires low-power consumption.**

Keywords—magnitude comparator, binary comparator, pass transistor logic, transmission gate, conventional static CMOS logic, gate diffusion input technique.

I. Introduction

Due to rapid elcalation in usage of portable devices, demand for high-performance circuits in electronic devices have become a crucial need [1-4]. Modern highly complex electronic devices requires ICs and processors that have high efficiency and speed [5-6]. Therefore, efficient design methologies are highly demanded in order to cope up with the expected performance parameters of modern devices [7-9]. In order to meet the performance requirements, many efficient design techniques have been developed and implemented [10-12].

MC in digital ALU is the component used for comparing two binary digits or numbers [13]. By comparing two binary numbers, it provides output if they are greater than or equal to one-another [14-15]. This component in ALU is highly utilized in parallel processing and digital signal processing [16-17]. In addition, instruction sorting in microprocessors requires MC [18]. Hence, because of having significant demand in VLSI design, an effective MC design will bring about great performance escalation of ALU.

In this research, a hybrid two-bit magnitude comparator using PTL, TGL and C-CMOS logic has been presented. Performance of the proposed hybrid two-bit MC has been compared with the existing designs to validate its effectiveness. Cadence design tools have been used in this research for performance evaluation in standard 90 nm technology node.

II. Literature Review of Two-Bit Magnitude Comparator Designs

Due to wide area of applications, MC has become a prime subject of interest for which numerus two-bit MC designs have been implemented by researchers [19-20]. Two-bit MC using PTL utilizes 40 CMOS transistors is reported in [21]. The major disadvantage associated with this sort of design is degradation of voltage since N-MOS only passes strong logic 0 (zero). Hence, PTL MC suffers from low driving power and has high power consumption.

Later on, in order to overcome the disadvantages of PTL, C-CMOS logic has been developed. This design employs complementary set of N-MOS and P-MOS transistors for full-swing logic implementation. Two-bit MC design using C-CMOS logic has been reported in [22]. Transistor count (TC) for this design is 66. This high TC makes the input impedance high which accounts for high delay. However, in case of driving power, the design is dignificantly robust.

TGL do not incorporate the weakness of voltage degradation unlike PTL. However, high TC in TGL results in high area in silicon chip. Moreover, driving power is also an issue in this sort of design. Two-bit MC using TGL is reported in [23] which employs 66 transistors.

Two-bit MC design employing Full Adder (FA) cell is reported in [24]. The design employed 30 transistors and used FA design in [25]. TC of this MC is low while compared to the previously mentioned PTL and C-CMOS MCs.

Gate Diffusion Input (GDI) method enables designers to build VLSI circuits with less TC compared to other design methods [26-27]. Major drawback of this design is the voltage degradation unlike PTL. Two-bit MC using GDI technique in [28] only requires 28 transistors which is the lowest TC reported in this research.

Hybrid logic style based two-bit MC in [29] requires 46 transistors. The design used hybrid logic based AND and XNOR gates for internal logic signal generation. Then the internal signals are send to C-CMOS based circuit for generating output terms. Major advantage of this design lies in the intelligent use of C-CMOS method based circuit in the outer signal generation [30].

Dynamic logic based MC designs are also available in literature [31-32]. However, the porposed and the other MCs mentioned are static CMOS based for which comparison in this research are conducted only among static CMOS based designs.

III. Proposed Design

To have descriptive analysis of proposed two-bit MC cell, design methodology has been described using the following sub-sections.

TABLE I. Table of Operation of a Two-Bit Magnitude Comparator

Inputs		Outputs		
A_1, B_1	**A_0, B_0**	**AGB**	**ALB**	**AEB**
$A_1 = B_1$	$A_0 = B_0$	0	0	1
$A_1 = B_1$	$A_0 > B_0$	1	0	0
$A_1 = B_1$	$A_0 < B_0$	0	1	0
$A_1 > B_1$	X	1	0	0
$A_1 < B_1$	X	0	1	0

AGB: A>B (A is greater than B)
ALB: A<B (A is Smaller than B)
AEB: A= B (A and B are equal)
X= Don't care condition

A. Understanding Logic Operation of Two-Bit MC:

In order to be able to design a two-bit MC, at first, it is necessary to fully understand its logic of operation. Operation technique of two-bit MC is conveyed using Table I. In Table I, the binary numbers required to be compared are denoted as A and B. The most significant bits are represented by A_1 , B_1. Least significant bits of the binary numbers A and B are denoted by A_0 and B_0 respectively.

Operation conveyed in Table I can be written using Boolean Algebra in order to express the operation of two-bit MC using Boolean Equations. The equations are

$$AGB = A_1 + A_0 XN_1 \quad (1)$$
$$ALB = B_1 + B_0 XN_1 \quad (2)$$
$$AEB = XN_0 XN_1 \quad (3)$$

Here, $XN_0 = A_0$ XNOR B_0 and $XN_1 = A_1$ XNOR B_1

Hence, it is clear from the above mentioned discussions that implementation of the two-bit MC would require AND and XNOR operations. Later, a complex logic gate would be required to implement Boolean equations stated in (1) and (2)

B. Block Diagram

As per observations made from Table I and eq. 1-3, proposed 2-bit MC operation process is expressed using Fig. 1. Initially, the inputs are passed onto the TGL and PTL based hybrid XNOR gates in order to generate the internal node signals. Usage of C-CMOS method in the initial circuitry is deliberately avoided in order to reduce the high-input impedance of C-CMOS design. In this way, the RC time constants of the circuits become low. Due to this low RC time constant, the input signals can quickly pass along the circuit components for which delay has been reduced significantly.

The output signals generated from the hybrid XNOR gates are later passed onto C-CMOS based circuit which provides full swing output signals having high driving power. C-CMOS logic is intentionally placed in generating the output signals in order to provide robustness and driving power to the circuit.

C. Circuit Implementation

Since Fig. 1 implies that MC implementation would require AND and XNOR gates, finding the perfect AND and XNOR gates would result in performance increment of the proposed two-bit MC. Various AND and XNOR gates are available in [33-39]. Logic gates in [33] don not provide full swing operarion. As per simulation results seen in [39], logic gates in [39] is highly efficient and would bring about great perfromacne enhancement if applied to the proposed MC. Hence, logic gates designed in [39] have been chosen in order to implement the porposed MC circuit diagram.

For AND operation, C-CMOS based design has been utilized. For ALB and AGB, XNOR operation is required. This XNOR gate can be designed by interchanging A and $\bar{A}$ in the XOR gate represented in [39]. After employing the cirucits for generating XNOR singnals, C-CMOS circuits has been developed for the Boolean expressions in (1) and (2). Full circuit diagram has been depicted in Fig. 2.

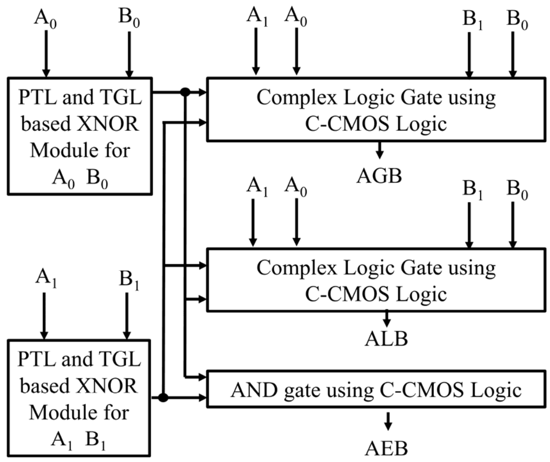

Fig. 1. Block diagram of proposed magnitude comparator.

IV. Simulation Result Analysis, Comparison and Discussion

In order to compared the porposed MC with the existing ones described in section II, circuit simulation has been conducted in 90 nm CMOS process. Cadence CAD tools have been utilized for this purpose. All designs have been simulated for 1.0 V supply voltage so that the designs get similar environment for comparison. Obtained simulation results have been presented in Table II and Fig. 3-5.

The simulation results in Table II states that the porposed MC is highly efficient in case of propagation delay. PTL based MC has quite satisfying speed. Compared to the other designs, GDI based MC in [28] has very high delay.

Power consumption of the design is also low compared to the designs in [21-23, 29]. Only MC design in [28] has less average power than the two-bit MC in this research work. However, there is a problem associated with GDI based MC in [28]. The design only consists of circuits for AEB and AGB. Circuit for ALB is missing in this design. Adding this circuit would add more transistor to the overall design and will result in increased power dissipation.

In case of PDP, porposed design achieved the best position which proved its effectiveness. Although having satisfying delay, PTL based design has the highest power dissipation for which its PDP is high. GDI based design obtained highest PDP due to its high delay.

V. Conclusion

This works presented the design of a two-bit MC. The proposed MC utilized PTL and TGL based logic circuits for generating the initial signals from the inputs which reduced TC. Output terminals consisted C-CMOS circuits to provide robustness and driving power. For comparison purpose, porposed MC has been simulated in 90 nm CMOS process along with the existing ones. The porposed MC achieved quite satisfying performance parameters which proved its effectiveness. Due to the high performance obtained by the porposed design, it is proved that the design is highly useful for designing modern microprocessors [40-43].

TABLE II. Simulation Result of Magnitude Comparators

Comparator logic style	Ref no.	TC	Power (c)	Delay (ns)	PDP (fJ)
PTL	[21]	40	11.031	0.287	3.166
C-CMOS logic	[22]	66	13.589	0.311	4.226
TGL MC	[23]	66	15.116	0.356	5.381
FA based MC	[24]	30	Failed		
GDI Method	[28]	28	7.783	0.513	3.992
Hybrid	[29]	46	7.802	0.212	1.654
Proposed	….	46	7.792	0.192	1.496

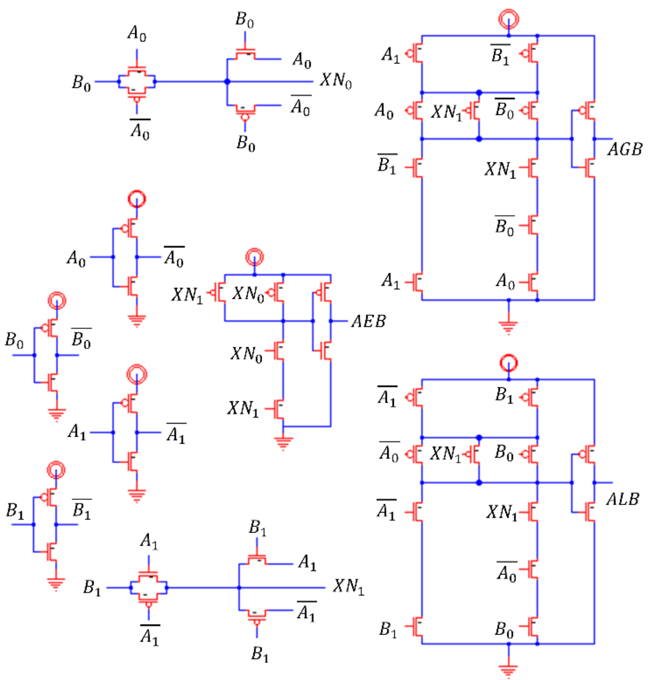

Fig. 2. Proposed two-bit magnitude comparator.

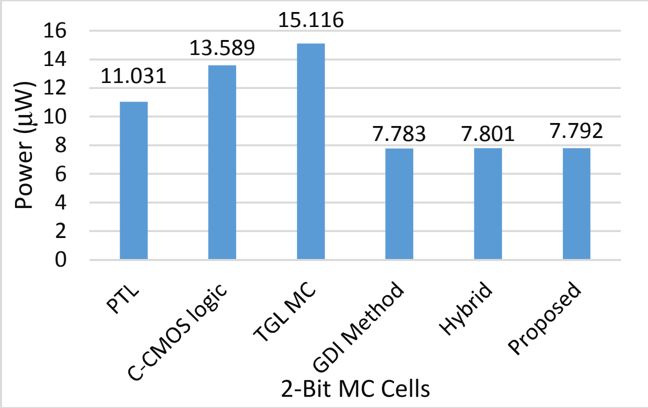

Fig. 3. Simulation results of power dissipation.

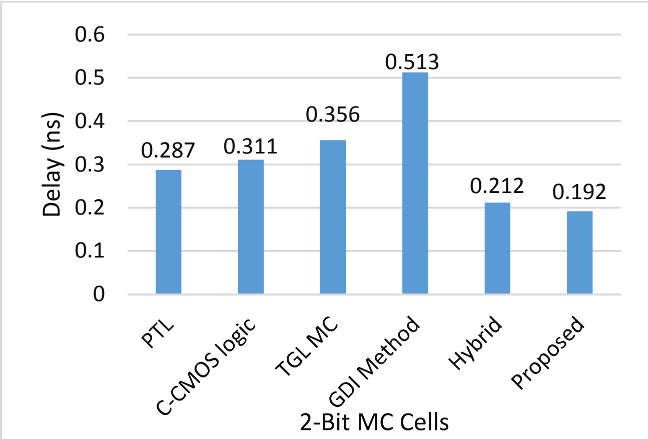

Fig. 4. Simulation results of propagation delay.

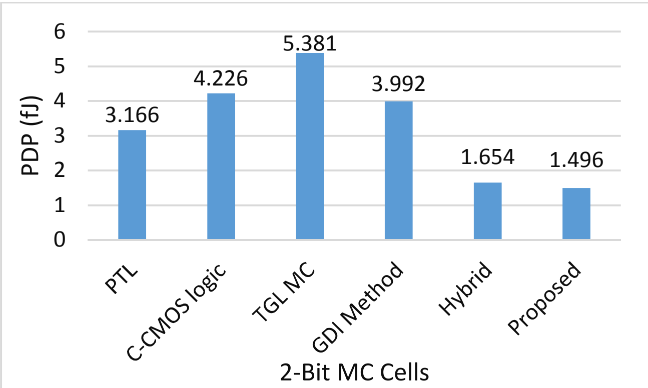

Fig. 5. Simulation results of PDP.

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